

Chapter 2 Addendum: Proposed Structural Constraints

Structural Closure

Every proposed modification should preserve a connected and internally consistent structural network.

Computational Trade-Off

Additional structural complexity should be balanced against available computational or descriptive capacity.

Semantic Relevance

Each node n is assigned a semantic relevance $V_r(n)$. Nodes with $V_r(n)=0$ are candidates for pruning.

Structural Pruning

Inactive nodes may be removed to improve structural efficiency while preserving explanatory capability.

Predictive Integrity

Dependency relationships should remain acyclic and computationally well-defined to avoid recursive ambiguity.

$$\begin{aligned} G &= (N, E) \\ G' &= (N \cup \{n_{\text{new}}\}, E') \\ \forall n \in N, \exists P(n, n_{\text{new}}) \\ C_{\text{used}} + C_{\text{new}} &\leq C_{\text{total}} \\ V_r(n) &> 0 \\ V_r(n)=0 &\Rightarrow \text{prune}(n) \\ N' &= N - \{ n : V_r(n)=0 \} \end{aligned}$$

Scientific Status: Proposed internal axioms of Universal Harmonic Theory requiring independent development and validation.